

Antonio Carmagnani

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Dear Hiring Team at Hines,

I trained as an architect, and the interest that has run alongside that from the beginning is data - using it to understand what has been built and what could be built better. That is what took me from architecture practice into feasibility and analysis, and then to a Master's Degree in Architecture Computation at The Royal Danish Academy, a programme built around using data and technology to improve the decisions made in real estate and construction. The through line has been consistent, which is that I would rather build the analytical tool that answers a question than argue the question without one. The Conceptual Construction Group asks for exactly that, a database assembled out of every European project and the tools that turn it into forward cost and programme forecasts the investment teams can act on.

My Master's Degree in Architecture Computation gave the analytical side of my work its proper depth. I built a predictive property valuation model using ensemble learning across XGBoost, LightGBM, Ridge and CatBoost that finished in the top two per cent of its competition, and a machine learning tool that generates site-specific development recommendations for buildings in Copenhagen by combining financial, social, sustainability and geospatial data, with an interface that lets non-technical stakeholders adjust the impact drivers and weightings themselves. The second of those mattered more to me than the first, because it forced the modelling to survive contact with people who would never open the code. Forecasting construction cost and programme from historic project data is that same exercise, and what carries over hardest is that the output has to be usable by the people it was built for.

At Isay Weinfeld Architects I wrote a Grasshopper plugin in Python that generates bespoke staircases directly from a floor plan, which took a task that had run to several days down to roughly twenty minutes and made a far wider range of geometry testable within a project timeline. Building it was the smaller half of the work - the rest was sitting with the colleagues who had been doing it by hand to establish what they actually needed, presenting the tool in a studio workshop, and supporting its adoption across more than ten projects afterwards, which taught me that a tool nobody trusts enough to use has not been finished. Driving construction data out of every project across a region rests on the same thing, because a repository is worth only what the project teams are willing to put into it.

Hines runs projects from pre-investment through to completion and operation across thirty countries, which means the data this role assembles has somewhere consequential to go. That is where I want to keep doing the work I find most worthwhile, which is turning large and awkward bodies of data into something a business can decide on.

Best,

Antonio Carmagnani